

8 Finance

8.1 Simple and Compound Interest

Definition 8.1.1 (Simple One-time Interest) The interest is only computed once. For example, a small loan which is paid back in full in three months, with some percentage of interest.

Go over derivation of formulas. We will use P_0 to denote the principal (the starting amount of a loan or account balance).

$$I = P_0 r$$

$$A = P_0 + I = P_0(1 + r)$$

Example 8.1.2 You lend \$300 dollars to a friend, who agrees to pay the amount back at the end of 3 months with 3% interest. How much will you make from this?

We can generalize the above idea pretty easily.

Definition 8.1.3 (Simple Interest Over Time) If the interest is repeatedly added after a certain amount of time, denoted t , and the amount of interest does not change, this is called simple interest over time.

Go over the derivation of formulas $I = P_0rt$, and

$$A = P_0 + I = P_0(1 + rt).$$

Example 8.1.4 Treasury Notes (T-notes) are bonds issued by the federal government to cover its expenses. Suppose you obtain a \$1,000 T-note with a 4% annual rate, paid semi-annually, reaching maturity in 4 years. How much interest will you earn?

$P_0 = 1000$, $r = 0.02$, $t = 8$ (8 6-month periods).

Also mention what an APR is here, since most of the examples in this chapter will involve this.

Definition 8.1.5 (Compound Interest) When the interest is reinvested and can itself earn interest, this is called compound interest.

Go over derivation of formula.

$$P_m = P_0 \left(1 + \frac{r}{k}\right)^n$$

where n is the total number of compounds, and k is the number of compounds per year. Mention that it is more common to use a formula involving the number of years.

Mention that this is the formula we use when we deposit money into an account and don't deposit any more, we just let it sit there.

Example 8.1.6 A certificate of deposit (CD) is a savings instrument that many banks offer. It usually gives a higher interest rate, but you cannot access your investment for a specified length of time. Suppose you deposit \$3000 in a CD paying 6% interest, compounded monthly. How much will you have in the account after 20 years?

Example 8.1.7 You open a checking account with an APR of 3.5% by depositing \$1500 at your local bank's branch office. Assuming you don't deposit more money into the account, how much will be in the account in 12 years if interest is compounded once a year? What if interest is compounded each quarter? How about each month?